

Systematic Review

Agile Practices and Agility in Non-Software SMEs: A Systematic Literature Review

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Abstract

(1) Background: Research on non-software small and medium-sized enterprises (SMEs) uses *agile* to describe named management practices, related approaches such as Lean, and organizational capabilities. Conflating these constructs obscures what firms actually do. (2) Methods: We revisited a systematic search of Scopus and Web of Science that identified 502 records and yielded an original set of 53 studies. We linked each record to its PDF and extracted study context, design, principal finding, and reported enabling or limiting factors. (3) Results: The map distinguishes 11 practice-focused records, 8 on adjacent approaches, 25 on agility as a capability, and 9 retained in a separate peripheral category. These are thematic strata of the historical set; the 44 non-peripheral records form the domain-focused synthesis. Some practice-focused records assess awareness, feasibility, or proposed models rather than observed adoption. (4) Conclusions: The literature supports a differentiated account of agile practices and agility in non-software SMEs; it does not establish a general causal effect of Agile Methodology adoption on SME performance. Historical screening gaps and the absence of an independently verified risk-of-bias appraisal limit certainty.

Keywords: agile practices; organizational agility; SMEs; systematic literature review; non-software; manufacturing



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1. Introduction

SMEs often face resource constraints when responding to volatile demand. Management practices that provide feedback and permit adaptation may be useful, but their effects depend on the firm's work and context [1,2].

The 2001 Agile Manifesto articulated values for iterative and adaptive software development [3]. Named methods and related practices have since been discussed outside software development. The word *agility*, however, is also used for an organizational capability, without evidence that a firm implemented any particular method. This ambiguity motivates the present review.

This study examines both explicit agile practices and the broader use of agility as an organizational capability in non-software SMEs. A study that measures strategic or

supply-chain agility without documenting an adopted method cannot establish adoption of Scrum, Kanban, or another named method. We therefore distinguish practices, adjacent approaches, and capabilities throughout the synthesis. The research questions are as follows:

- **RQ1.** Which organizational areas of non-software SMEs are examined in studies of agile practices and agility?
- **RQ2.** Which explicit agile practices, adjacent approaches, and agility capabilities are examined?
- **RQ3.** Which drivers and barriers are reported for these practices and capabilities?

The original 53-record set spans manufacturing, textiles, services, supply chains, and other settings. Its conceptual diversity is itself a result: only a subset examines explicit agile practices, whereas many studies measure agility as a capability. We therefore interpret reported flexibility, efficiency, and responsiveness in relation to the construct and study design that generated each claim.

The original systematic search of Web of Science and Scopus identified 502 records and retained 53 after screening. We audit that historical set against the available full texts and describe the screening evidence and its limits in Section 3; the new construct codes identify records whose final eligibility remains uncertain.

The article is organized as follows. Section 2 presents the key concepts necessary to follow the course of this research. Section 3 explains and applies the research methodology used in this study. Section 4 discusses the results obtained. Section 5 briefly presents a future research agenda, and Section 6 concludes and describes the limitations of this research.

Data Access of SLR: The list of selected papers in each stage of this systematic literature review is available on <https://tinyurl.com/paper-agile-slr> (accessed on 28 February 2025).

2. Background: Agile Practices and Organizational Agility

This section defines the constructs used in the review. A named agile method specifies recognizable roles, events, artifacts, or work practices that can be observed in a firm. An adjacent approach may share iteration or continuous improvement without being a software-origin agile method. Organizational agility denotes a firm's capacity to sense and respond to change; a study may measure this capacity without observing any particular agile method. These distinctions are coding rules, not claims that the three constructs are interchangeable.

2.1. AMs from Traditional Business Management Methodologies

Organizations require principles and practices that enhance productivity and support effective decision-making, particularly in uncertain and rapidly evolving environments. Selecting an appropriate management methodology is essential, as it directly impacts the achievement of organizational performance indicators.

Until the 1990s, most industries relied on traditional management methodologies characterized by structured and planned approaches [4,5]. These approaches emphasized standardized processes, extensive documentation, and precise planning to ensure control and predictability [6].

In the software industry, traditional methodologies such as the *Waterfall Model* and *Evolutionary Development Models* were widely applied, especially for large-scale projects [7]. The Waterfall Model introduced by Royce in 1970 defined a sequential process comprising five stages: (1) requirements analysis and definition, (2) system design, (3) implementation and unit testing, (4) system integration and testing, and (5) operation and maintenance. Although this model offers comprehensive documentation and integration with engineering models, it lacks flexibility for adapting to changing requirements.

In contrast, the Evolutionary Development Model follows an iterative approach where early system versions undergo user testing and feedback. Based on these evaluations, new versions are developed until the system meets the defined requirements. Unlike the Waterfall Model, it acknowledges that requirements may evolve during the software lifecycle [7].

Different approaches assign different roles to planning, feedback, and adaptation. These are conceptual distinctions; the present review tests whether an included study actually describes a practice before treating it as evidence of method adoption.

2.2. Agile Methodologies in the Software Development Industry

The 2001 Agile Manifesto articulated values associated with iterative and incremental software development [8]. Individual methods have distinct histories and operational components; the manifesto's date is not the inception date of every method.

The Agile Manifesto establishes a conceptual framework based on four fundamental values and twelve principles, which guide adopting this work philosophy in software project management. These values, represented in Figure 1, promote a new organizational mindset centered on flexibility, collaboration, and the continuous delivery of value [8,9].

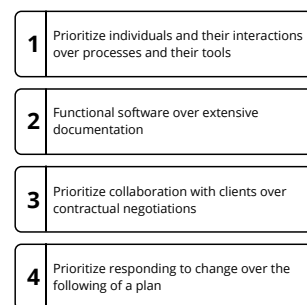


Figure 1. Values of the Agile Manifesto.

The twelve principles, illustrated in Figure 2, describe how these values translate into concrete practices within agile teams. These principles emphasize the importance of customer satisfaction, the continuous delivery of functional software, adaptability to change, and continuous improvement, among other key aspects of agile software development [10].

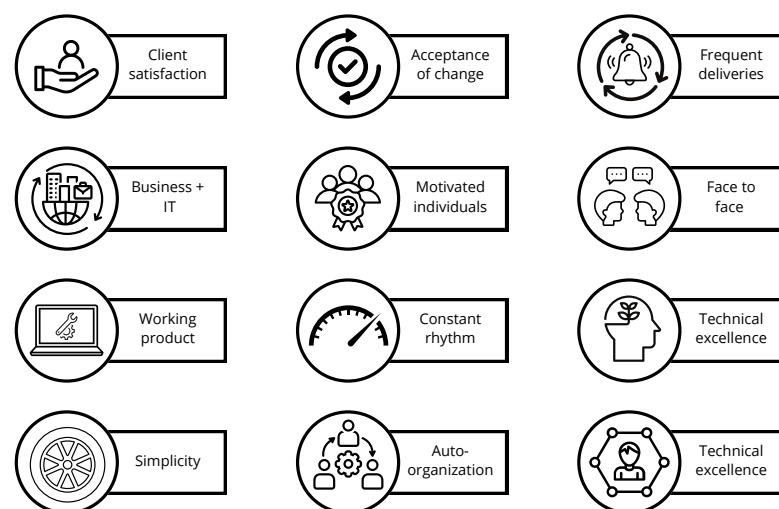


Figure 2. Principles of the Agile Manifesto.

AMs emerged to address the limitations of traditional software development approaches. While conventional methodologies emphasize exhaustive planning and rigid

change management, AMs introduce a paradigm shift by prioritizing adaptability, iterative progress, and stakeholder collaboration. This transition has enabled organizations to achieve shorter delivery cycles, cost reductions, simplified documentation processes, and higher customer satisfaction [9].

Over the past decade, AMs have become one of the most widely adopted development methodologies in the software industry [11]. Their success stems from their ability to enhance responsiveness to evolving market conditions while fostering continuous improvement.

Table 1 presents a comparative analysis of traditional and agile methodologies, highlighting the fundamental differences in their principles and implementation. While traditional approaches prioritize meticulous planning, contractual stability, and large team structures, agile methodologies emphasize flexibility, iterative development, and active customer involvement. For example, in a traditional manufacturing company, strict contracts and detailed planning might lead to delays if unexpected supply chain disruptions occur. In contrast, an agile approach allows for rapid adjustments, enabling teams to reallocate resources and modify production strategies dynamically.

Table 1. Methodologies comparison.

Traditional Methodology	Agile Methodology
Sequential stages	Iterative increments
Fixed scope and formal change control	Regular reprioritization
Function-specific teams	Cross-functional collaboration
Up-front planning	Frequent feedback
Limited customer involvement	Regular stakeholder feedback

Types of AMs

Various AMs have emerged over time, each with distinct approaches and characteristics. However, they all share fundamental principles such as incremental and iterative development, rapid adaptation to change, customer satisfaction as a priority, and a commitment to delivering high-quality products.

Among the named methods is Scrum, which organizes work into iterative cycles (Sprints). Extreme Programming (XP) emphasizes software engineering practices such as pair programming and automated testing. Table 2 lists conceptual reference points for coding. Inclusion in this background table does not imply that any of the 53 studies implemented the method.

Table 2. Named agile methods used as conceptual reference points.

Agile Method	Description	Reference
Crystal Methodology	Designed for small teams, focused on communication and collaboration. Based on individuals rather than structured processes.	[12]
Scrum	An iterative methodology that divides work into Sprints. It is organized into roles (Scrum Master, Product Owner, Developers), artifacts (backlogs, increments), and events (Daily Scrum, Sprint Review, etc.).	[13–16]

Table 2. Cont.

Agile Method	Description	Reference
Extreme Programming (XP)	Centers on short iterations and continuous customer feedback. Its goal is to minimize costs associated with unexpected changes.	[17–19]
Kanban	A visual method for managing workflow and work in progress.	[14,15]

Design Thinking is a user-centered innovation approach [20–22], and Lean Startup is an experimentation approach [23]. Neither is classified here as an Agile Methodology solely because it uses iteration. Lean manufacturing and Lean construction are similarly coded as adjacent operational approaches. In the included-study audit (Appendix A), an explicit agile-practice code requires a named method or identifiable agile project/production practices in the study itself; capability studies receive a separate code. This rule prevents an agility outcome from being counted as evidence that a firm adopted Scrum or another named method.

2.3. AMs in Business SMEs

While AMs are widely implemented in the software sector, large enterprises in other industries, such as technology and finance, have adopted them as well. Examples include Google, Amazon, BBVA, and Microsoft [8]. These transitions often involve changes in organizational structure, team roles, and work processes [24].

Non-software SMEs are studied in manufacturing, supply-chain management, services, and other activities. The study-level map distinguishes reports of implemented project or production practices from measures of supply-chain or organizational agility; the latter cannot by themselves establish method adoption (Appendix A).

However, SME adoption is influenced by unique characteristics. On the one hand, SMEs benefit from flatter hierarchies and greater structural flexibility [5]. On the other hand, they may face cultural resistance, fear of change, and limited resources [6,25]. A lack of awareness or understanding of AMs can also hinder implementation. Understanding these dynamics is essential for promoting effective and sustainable adoption of agility in non-software SMEs.

In this study, the term *business SMEs* refers to non-software SMEs, i.e., small and medium-sized enterprises whose core activity is not software development, as opposed to software-focused firms, which are excluded from this review under criterion EC3, defined in Section 3.3.

3. Methodology

This section describes the methodology applied in this study, which follows the guidelines proposed by Petersen et al. [26] for conducting Systematic Literature Reviews (SLR). This methodology, complemented by the works of Petersen et al. [27], Wohlin [28], and Petticrew and Roberts [29], provides a structured framework for performing a comprehensive and organized analysis of the study area.

This review is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [30]. The completed PRISMA 2020 checklist is provided as Supplementary Material, and Figure 3 presents the study selection flow in PRISMA style. Because this article revisits a historical search, some items could not be fully reported; these are detailed in the checklist.

This section is divided into six subsections that detail the steps of this methodology: (1) Research Questions, (2) Search, (3) Study Selection, (4) Data Extraction, (5) Analysis and Classification, and (6) Validity Assessment.

3.1. Research Questions

This review examines agile practices and agility-related capabilities in non-software SMEs. The three research questions in Table 3 separately address organizational area, focal construct, and reported drivers or barriers.

Table 3. Research questions.

ID	Research Question
RQ1	Which organizational areas of non-software SMEs are examined in studies of agile practices and agility?
RQ2	Which explicit agile practices, adjacent approaches, and agility capabilities are examined?
RQ3	Which drivers and barriers are reported for these practices and capabilities?

3.2. Search

After defining the research questions, we searched Scopus and Web of Science (WoS). The general terms combined agile/agility, SME variants, and business/management:

("agile" OR "agility") AND
 ("sme" OR "smb" OR "small and medium-sized enterprises" OR "small and medium-sized businesses") AND
 ("business" OR "management")

The databases were selected for their coverage of management and operations research. Database indexing is not a study-level quality appraisal. The original search was conducted in 2024; only complete publication years through 2023 were intended to be eligible. The working files do not record the exact execution date or a contemporaneous pilot-search log, so pilot sensitivity cannot be reconstructed as a verified historical procedure.

Table 4 reports the archived initial database queries and their returns (288 from Scopus and 214 from WoS). Both initial queries already excluded the databases' Computer Science category. This subject-category filter is distinct from the later study-level EC3 decision about whether a retrieved paper focuses on software development. The archived refined queries, including year, document-type, and language filters, are preserved in the companion workbook, sheet "Search Strings."

Table 4. Exact database-specific search strings.

Database	String	Records
Scopus	TITLE-ABS-KEY(("agile" OR "agility") AND ("sme" OR "smb" OR "small and medium-sized enterprises" OR "small and medium-sized businesses") AND ("business" OR "management")) AND (EXCLUDE(SUBJAREA, "COMP"))	288
WoS	TS=("agile" OR "agility") AND TS=("sme" OR "smb" OR "small and medium-sized enterprises" OR "small and medium-sized businesses") AND TS=("business" OR "management") NOT WC=(Computer Science)	214

Applying each platform's native filters for publication year (2010–2023), document type (article or review), and language (English) narrowed 502 records to 238 (113 from Scopus and 125 from WoS). The original files preserve these queries and counts but no pilot results, test set, or exact search date. The requirement for an SME term may have missed studies that describe small firms using different terminology; we did not rerun an expanded search for this revision.

3.3. Selection

The articles were selected in several stages, applying a systematic approach to ensure the relevance and quality of the included studies. Figure 3 summarizes the number of studies retained at each stage. The stages are described as follows:

- **Application of Inclusion and Exclusion Criteria:** Starting from 238 records after the database-native year/type/language filters, duplicates were removed (56 by title, 12 additional by DOI), reducing the sample to 170. Study-level criteria EC3 (software-development focus) and EC4 (secondary systematic reviews or mappings) were then applied together, reducing the sample to 165. The historical sheet does not separate the five exclusions by criterion.
- **Title and Abstract Screening:** The original review screened the 165 remaining records for relevance to SMEs and agile practices or agility, applying IC4 and EC2. This step narrowed the set to 85, but the original file does not record a reason for each of the 80 exclusions.
- **Full-Text Analysis:** The original team reviewed the introductions and conclusions of the 85 selected articles and other sections when necessary. This stage retained 48. The workbook flags 35 of the 37 records not carried forward as rejected and gives a brief free-text note for each of the 37; these notes are not coded by exclusion criterion.
- **Snowball Method:** To minimize the risk of missing relevant studies, we applied the snowball method following the guidelines of [28]. This approach identified three additional articles, resulting in 51 articles.
- **Review of Excluded Articles:** The original review reinstated two records, producing the historical set of 53. The workbook identifies them but does not record why they were reinstated.

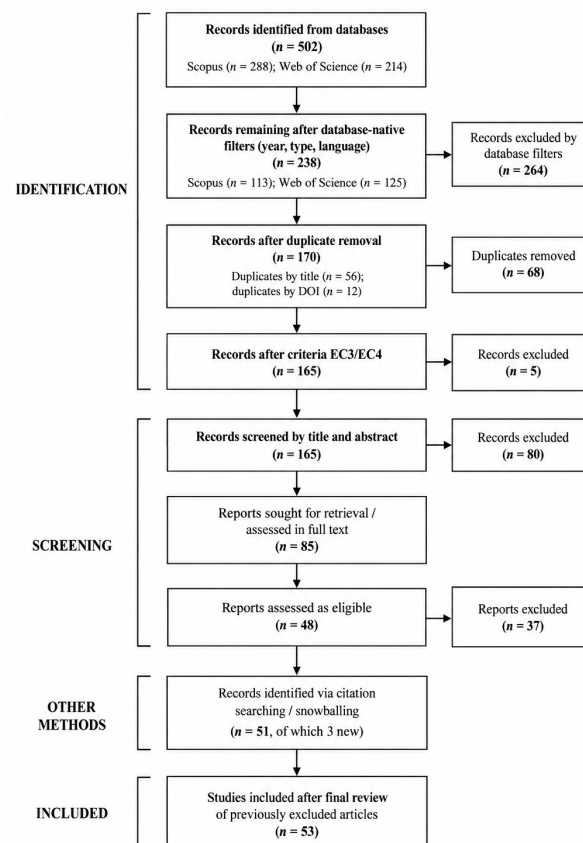


Figure 3. Archived selection flow for the original 53-record set.

Table 5 lists the full set of inclusion (IC) and exclusion (EC) criteria applied at each stage above.

Table 5. Inclusion (IC) and exclusion (EC) criteria applied during study selection.

Type	Code	Criterion
Inclusion	IC1	Language: include only English
Inclusion	IC2	Date: include studies published from 2010 to 2023
Inclusion	IC3	Database document type: article or review (before study-level EC4 screening)
Inclusion	IC4	SME study relevant to an explicit agile practice, an adjacent operational approach, or agility as a capability
Exclusion	EC1	Duplicate documents across search engines or in content; the first occurrence is kept and the rest excluded
Exclusion	EC2	Documents with no established keywords, or that do not address the research questions
Exclusion	EC3	Study centered on software development rather than a non-software SME (after the database subject-category exclusion)
Exclusion	EC4	Secondary systematic reviews or systematic mappings
Exclusion	EC5	Publications not in English

3.4. Data Extraction

A structured template was used for data extraction. This template allowed us to uniformly organize the extracted information, facilitating the analysis and subsequent classification of the selected studies. The fields used in the template are as follows:

- **General Study Information:** ID, Title, Reviewer, Database, Year, Source Title, Document Type.
- **Author Information:** Authors, Authors' Keywords.
- **Article Content:** Abstract, Abstract/Introduction, Full Text.
- **Technical and Relevance Details:** DOI, Software, Systematic Literature Review (SLR).
- **Research Questions:** RQ1, RQ2, RQ3.
- **Additional Observations:** Observations.

3.5. Analysis and Classification

For this revision, we reconciled each of the 53 original records with one accessible PDF and extracted the focal construct, study country when stated, industry, SME context, study design, principal finding, and factors reported as enablers or barriers. We checked the abstract and relevant methods, results, and discussion passages in the supplied PDFs; the record-level extraction and source notes are in `data/study_evidence.csv`, with compact tables in Appendix A. We used four mutually exclusive primary construct codes: *direct* for a study centered on an explicit agile method or identifiable agile project/production practice, *adjacent* for an operational approach such as Lean that shares some principles but is not itself evidence of Agile Methodology adoption, *capability* for agility measured or theorized as an organizational property, and *peripheral* when the article does not establish an agile practice or agility as a central research construct. When a paper discussed more than one construct, the code describes its focal research object. A separate evidence-type field distinguishes observed implementation, survey association, proposed model, and peripheral publication; a *direct* code alone does not show that a method was adopted. Following the authors' scope decision, the nine peripheral records remain a separately labeled stratum of the historical 53 and are not used to infer method adoption. The 44 non-peripheral records form the domain-focused synthesis. Lean remains an adjacent approach.

RQ1 uses the primary organizational area recorded for each paper; industry and study country are separate context fields. RQ2 uses the construct code, focal practice/capability, and evidence type. For RQ3, one paper may contribute to several factors. The newly ex-

tracted reported enablers and barriers are separate from the historical citation-to-category links in Tables 6 and 7. We linked those citations back to study records by DOI or bibliographic title; all cited keys resolved, but this verifies attribution, not the semantic correctness of the old category assignment. The historical review did not preserve independent code-level decisions or a formal codebook, so the old categories remain descriptive.

Table 6. Themes originally coded as drivers of agile practices or agility.

Driver	References
Organizational structure and governance	[31–39]
Leadership and strategy	[5,6,40,41]
Organizational culture and learning	[24,42,43]
Market dynamics and environment	[6,33,40,44–49]
Customer and client focus	[33,44,47,49–51]
Innovation and technology	[33,34,36,42,43,45,52,53]
Resource management	[5,32,33,39,54–56]
Agility and flexibility	[40,56–59]
Partnerships and relationships	[39,51,60,61]
Process optimization and supply chain	[37,55,61–65]
Capabilities and synergies	[31–34,36,44,45,54,56,57]
Industry and market-specific requirements	[48–50,52,66]

Table 7. Themes originally coded as barriers to agile practices or agility.

Barrier	References
Resistance to change	[31,32,36,47,49,50,59,62,66–69]
Resource limitations	[31,36,37,51,52,54,57,58,64–66]
Training deficiencies	[32,49,59,62,66,67,69]
Cultural barriers	[31,61,62,70]
Technological challenges	[31,47,48,50,52,57]
Leadership and management issues	[36,46,56,67,70]
Financial constraints	[31,42,50,55,64,66,68]
Communication issues	[51,61,62,71]
Market uncertainty	[48,54,56,63,66]
Collaboration and coordination challenges	[48,60,61,68]
Integration issues	[6,61,62]
Strategic planning and knowledge gaps	[5,39,46,47]

3.6. Validity Assessment

An earlier version described a collaborative review with consensus discussions. The preserved extraction workbook, however, contains one assigned reviewer per record and no record-level history of independent second or third judgments, disagreement resolution, or item-level coding. Consequently, we cannot verify the earlier statement that every article was independently cross-validated by at least three authors, and we cannot calculate inter-rater agreement from the available material. The revision adds a provisional, single-reader reporting appraisal for all 53 PDFs: SME context, sample or case description, data source, analysis procedure, and limitations are each marked clearly reported (Y), partly reported (P), not reported (N), or not applicable (NA). The record-level codes and rubric are supplied with the evidence CSV. They describe reporting completeness, not risk of bias or causal strength; no numerical quality score or exclusion threshold was imposed. A second independent appraisal is still needed before ranking evidence strength. We therefore qualify causal and managerial conclusions by design and observed versus proposed practice.

4. Results

4.1. Bibliometric Overview of the Included Studies

Before answering the research questions, we present a bibliometric overview of the *original* 53-record set (37 attributed to Web of Science and 16 to Scopus in the selection workbook). The unit is one retained record, with the year, source title, authors, and author keywords from that workbook. Annual production groups records by the workbook's year field; recurring outlets are counted by exact source-title strings; and the keyword network uses author keywords occurring in at least two records and links terms that occur in the same record. The co-authorship summary connects authors who share a record. Author affiliation countries were recovered from the original export records for 44 records and are distinct from the countries of the firms studied. Because the nine peripheral records remain in this historical set as a separately labeled stratum, these descriptive figures should not be interpreted as bibliometrics of the 44-record domain-focused synthesis. The figure-generation code and a complete normalization log for authors and keywords were not preserved, limiting exact regeneration of the network figures.

The 53 records have workbook year values from 2010 to 2023 and list 133 distinct authors (2.6 authors per record, on average) across 48 source outlets. Figure 4 shows their annual distribution; 31 of 53 (58%) carry a year of 2020–2023. This trend describes the original search result, not a measured growth rate of method adoption. One record entered as 2023 early access (study 35, Appendix A) appears in a journal volume dated 2024; its treatment depends on whether online-first or issue year governs the eligibility cutoff.

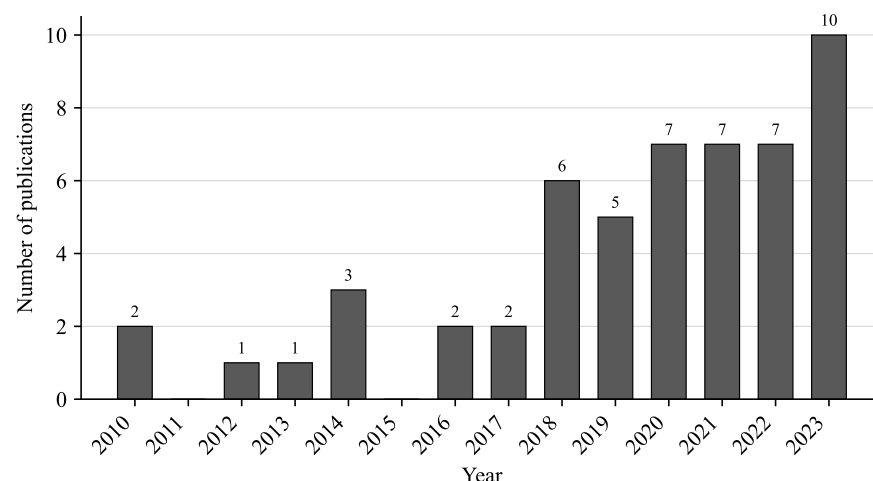


Figure 4. Annual scientific production of the 53 included studies (2010–2023).

Regarding publication venues, only 4 of the 48 sources contributed more than one record, while 44 sources (92%) contributed one record each. Figure 5 lists the four recurring sources, led by the *South African Journal of Business Management* (three records). This distribution concerns the mixed original corpus and cannot establish that research on a specific named methodology is fragmented.

We further examined the institutional affiliations reported for each author. This measure reflects where the *authors* are based, not the location of the SMEs studied. Country information could be recovered for 44 of the 53 records (83%); the remaining 9 lack suitable export metadata. These 44 records draw on author affiliations from 32 countries. Figure 6 shows the 15 most represented countries: the United Kingdom (6 records) and China (5 records) lead the list. Twelve of the 44 records with country data (27%) involve authors affiliated with more than one country. We do not infer geographic coverage of SME settings from these affiliation counts.

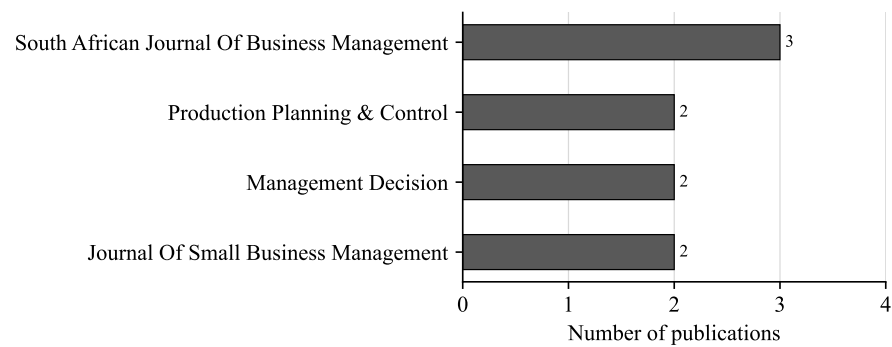


Figure 5. Source outlets with more than one included publication.

We also examined co-authorship patterns among the 133 authors of the 53 included studies. The resulting co-authorship network is highly fragmented: it forms 49 disconnected components (i.e., separate author teams that never share a co-authored study with another team in the sample), and the largest component includes only six researchers. In practice, this means that, with few exceptions (e.g., “Liu, HM” and “Yang, HF”, who co-author two of the included studies), each of the 53 studies is produced by an independent author team rather than by a small number of recurring research groups; we therefore report this fragmentation as a summary statistic rather than as a network diagram, since a plot of 49 mostly disconnected clusters would not convey additional information beyond the figures above.

Figure 7 presents the author–keyword co-occurrence network, restricted to the 29 keywords appearing in at least two original records. Nodes are sized by frequency and edges by record-level co-occurrence. Operations and supply-chain terms appear alongside organizational-capability terms. This is consistent with the study-level coding, which places 25 records in the capability class and 11 in the explicit-practice class; keyword proximity itself does not demonstrate implementation of a named method.

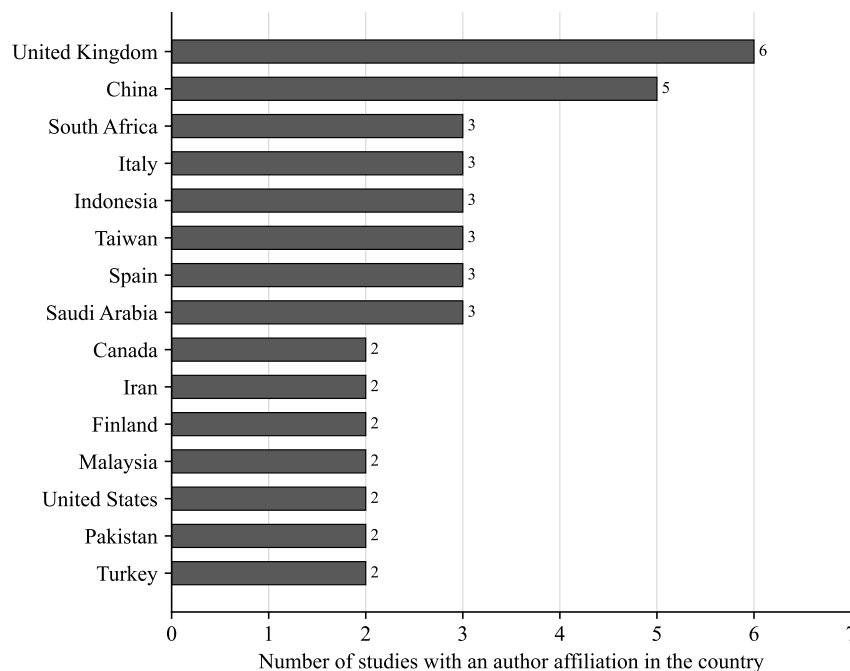


Figure 6. Countries of author affiliation (top 15) among the 44 included studies with extractable country data.

4.2. Answering RQ1 and RQ2: Organizational Areas and Focal Constructs

The 53-record evidence map (Appendix A) identifies each record's primary organizational area and separately reports study country, industry, SME context, design, and principal finding in Table A2. The domain-focused synthesis comprises 44 non-peripheral records; the other nine remain a visible separate stratum, preserving the historical PRISMA count. Project and operations management, supply chains, innovation, and workforce management all occur, but an industry label such as manufacturing must not be substituted for the function being studied.

The 11 direct records are heterogeneous. One centrally describes Scrum implementation in a UK bed-manufacturing SME [71]; four concern agile project management or a Lean-agile project office, two agile manufacturing, one Lean-agile garment operations, one general agile processes, one a proposed people-management model, and one the potential use of agile methods in construction. No record centrally studies Kanban or XP adoption in a non-software SME. Awareness surveys, feasibility judgments, modeled associations, and conceptual proposals within the 11 records are not counted as observed implementations. The eight Lean-related records remain adjacent approaches. Among the 25 capability records, the construct ranges from supply-chain and workforce agility to strategic and market agility; these studies do not document adoption of a named method merely by measuring agility. The exact study-to-construct links and evidence types are in the companion CSV.

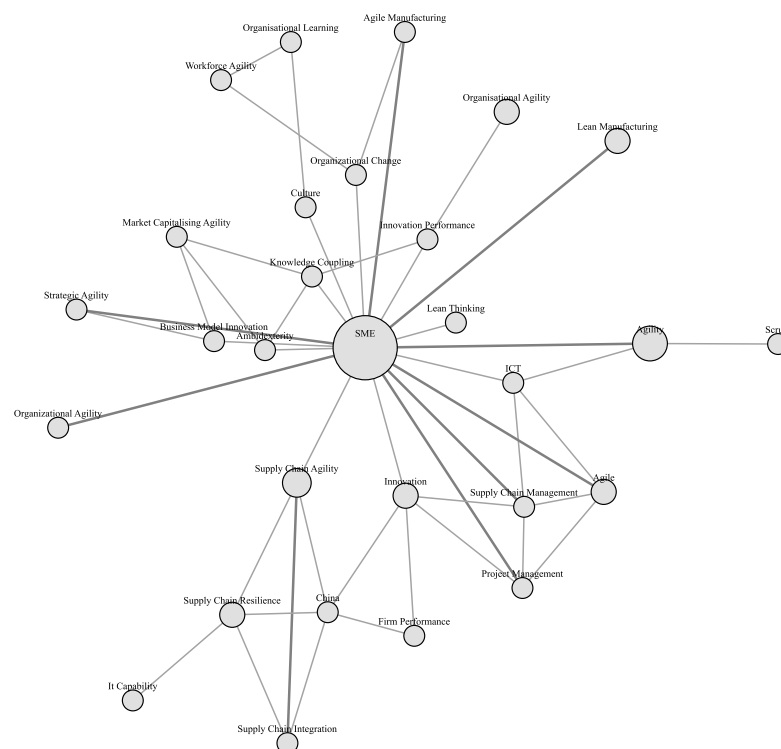


Figure 7. Author-keyword co-occurrence network (keywords appearing in at least two of the 53 included studies).

4.3. Answering RQ3: Reported Drivers and Barriers

Tables 6 and 7 retain the original review's thematic links to all 53 historical records for audit, including citations to peripheral papers. The domain-focused interpretation concerns the 44 non-peripheral records and uses the newly extracted factors in the companion CSV. The historical labels cover both implementation experiences and correlates of measured

agility; a citation is not proof that a specific named method was implemented. For example, Sukwadi et al. [31] study Lean–agile operations and supplier partnerships in garment supply chains, whereas Zhuo et al. [61] study supply-chain integration and resilience in China’s pig sector with agility as a mechanism. The latter does not test an Agile Methodology from Table 2. The study-level map allows readers to interpret each citation in its actual construct class.

The new study-level extraction also exposes variation hidden by the historical categories. In Thai SME exporters, perceived technology advantage supported adoption of an inventory system while perceived complexity reduced it [72]; this concerns a marketing-agility capability rather than method adoption. In two Swedish textile SMEs, Lean management changes improved coordination, but work stress increased in one case and decreased in the other [32]. In Indonesian garment SMEs, supplier partnership and agile supply-chain constructs were associated with performance, whereas the modeled leanness construct was not [31]. These contrasts show why a single list of generic “agile adoption barriers” is insufficient. The CSV preserves the newly extracted factors separately from the original table assignments, so readers can inspect the source-specific interpretation.

Table 7 similarly records the original barrier categories. A paper may appear in several categories. In particular, seven cited records appear in “Training deficiencies”; this is a descriptive frequency within the original 53-record set, not an estimate of the causal effect of training. Because independent code-level records were not retained, the assignments require a future second-coder audit before they can support strong frequency comparisons.

4.4. Integrative Interpretation of the Evidence

The study map supports a differentiated interpretation of how agility-related factors may be connected. Market uncertainty, customer requirements, leadership, resources, and learning are reported as possible *antecedents*. Explicit practices such as Scrum or agile project management, adjacent Lean practices, and coordination or knowledge-sharing processes are different possible *mechanisms*. Industry, SME size, supply-chain position, and available expertise are plausible *contextual conditions*. Flexibility, responsiveness, innovation, and performance are *outcomes reported or measured* in the original studies. These elements should not be collapsed into a single “adoption leads to performance” claim: in capability studies, agility is often itself a measured mediator or outcome, whereas in practice studies a specific work method can be observed. For example, the Scrum case [71] describes an implementation, while the strategic-agility study [41] analyzes a capability in business-model innovation. This framework organizes the heterogeneous findings; it is an interpretive synthesis, not a causal model tested across all 53 records.

5. Future Research Agenda

The following research agenda emerges from the results and analysis presented in response to the RQs of this paper (Section 4). Based on these insights, we identify critical areas where future research can deepen our understanding and provide potential and beneficial actions for SMEs operating outside the software industry. The agenda focuses on three main dimensions: developing sector-specific agile frameworks, identifying strategies to address the interplay of adoption drivers and barriers, and expanding research into under-explored regions and economic sectors.

5.1. Context-Specific Agile Frameworks for SMEs

Future studies should test how specific named practices operate in non-software SMEs and distinguish those practices from adjacent Lean approaches and measured agility capabilities. Comparative designs can examine whether a practice fits production, project,

service, or supply-chain work and document the resources required to use it. The current evidence map identifies these settings but does not establish which practice works best in each one.

5.2. Strategies to Overcome Barriers and Leverage Enablers in Agile Adoption

The descriptive categories suggest candidate relationships among leadership, learning, resources, resistance, and outcomes. They do not establish a tested causal pathway across the corpus. The framework in Section 4.4 specifies relationships that future intervention or comparative studies could examine.

Future research should aim to achieve the following:

- Develop and test interventions using a named agile practice while documenting costs, skills, and implementation fidelity.
- Conduct empirical studies or controlled pilots to evaluate the effectiveness of enabling factors (e.g., leadership engagement, strategic partnerships, or organizational learning) in counterbalancing resistance and resource limitations.
- Test multilevel models that distinguish a work practice from individual, organizational, and environmental conditions and from the measured outcome.

5.3. Explore Under-Studied Regions and Sectors

Future reviews should code the country where each SME was studied. The present affiliation analysis locates authors, not study settings, and therefore cannot justify a claim about which regions lack SME evidence. New primary studies should report their organizational and national context sufficiently to permit comparison.

Cross-sector studies should compare how specific agile practices are adapted and evaluated, including sectors that are sparsely represented in the original set. The reviewed data do not support a ranked list of neglected sectors because industry was not uniformly coded in the historical extraction file.

6. Conclusions

This review revisits 53 records originally retained from a search of non-software SMEs. The domain-focused synthesis comprises 11 practice-focused records, 8 adjacent approaches, and 25 agility-capability records; nine peripheral records remain visible as a separate stratum of the historical set. The study-level extraction now documents context, design, principal finding, and reported factors for every record. Practice-focused records include proposals, awareness surveys, and feasibility studies as well as observed implementations. The varied designs do not establish a general causal performance effect of agile adoption.

The original thematic tables identify leadership, learning, resources, cultural resistance, and training as recurring topics. The new extraction records article-specific findings and disagreements, including different stress outcomes across two Lean textile cases. Historical category frequencies describe citations in the original set; they are not pooled effect estimates or independently reconciled codes. The integrative interpretation in Section 4.4 offers a structure for future tests rather than a validated causal model.

From a practical standpoint, this study offers several implications for SME managers and decision-makers:

- **Specify the intervention.** Managers should document the named practice, the area of use, and the intended outcome before interpreting results. Evidence about supply-chain agility as a capability cannot establish the effects of Scrum or Design Thinking.
- **Assess the local context.** Resources, leadership, coordination, and staff skills recur in the reviewed studies, but their relative importance has not been compared across sectors in this corpus.

- **Treat training as a hypothesis to test.** Seven of the 53 original records were cited in the historical “Training deficiencies” category (13%; Table 7). This frequency motivates attention to skills but does not prove that a specific training intervention improves adoption or performance.

These implications are deliberately conditional. A manager or policymaker should select a practice on the basis of the firm’s work and evaluate it locally rather than treating the mixed literature as a universal implementation guide.

Limitations

The historical search covered Scopus and Web of Science but not gray literature [73]; its required SME term and software subject-category exclusions could have omitted relevant non-software studies. No contemporaneous pilot log or exact search date was preserved. The selection workbook retains the surviving records and stage totals; it gives no reasons for the 80 title/abstract exclusions or the two final reinstatements, and its notes on the 37 full-text exclusions are free text not coded by criterion. The nine peripheral records are retained as a separate stratum, so historical bibliometrics of 53 and the domain-focused synthesis of 44 have different denominators. The count refers to publications, not necessarily independent datasets: two Taiwanese papers by the same authors each report 260 SME responses, and possible sample overlap remains to be checked. One 2023-online/2024-volume year ambiguity remains. The original file has one assigned reviewer per record, no independent coding or consensus log, and no inter-rater statistic. The new five-item reporting appraisal is provisional and single-reader; it is not a validated risk-of-bias assessment. Some study contexts and SME characteristics are incompletely reported in the source PDFs. Bibliometric normalization and figure-generation code were not preserved. Study designs, sectors, and agility constructs vary substantially, precluding a pooled causal conclusion.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/app16199842/s1>, The completed PRISMA 2020 checklist.

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Appendix A. Study-Level Evidence Map

The table reports the focal construct and study design recorded in the source corpus. The categories are descriptive coding of the original 53-record set, not a retrospective claim that every record implements a named Agile Methodology. Direct means an explicitly named agile practice or method; adjacent means a related operational approach such as Lean; capability means agility assessed as an organizational property; peripheral means that the abstract does not establish an agile practice or agility as a central construct. Eligibility flags require author adjudication before a revised final sample can be asserted. For complete bibliographic metadata and the matching PDF filename, see the accompanying data/study_evidence.csv.

Table A1. Study-level map of the original 53 records.

ID	Study (Short Title)	Class	Focal Construct	Design
1	A case study on performance assessment. . .	direct	Agile processes	Case study
2	A model for digital development of an. . .	peripheral	Digital development model	Model development
3	Alternative Models of Demand-driven Supply. . .	capability	Supply-chain agility	Empirical study
4	Application of lean manufacturing concept. . .	adjacent	Lean manufacturing	Case study
5	Assessing the awareness of the agile. . .	direct	Agile manufacturing	Empirical investigation
6	Contextual Ambidexterity: Tackling the. . .	capability	Contextual ambidexterity	Multiple case study
7	Critical success factors of lean. . .	adjacent	Lean manufacturing	Ranking model
8	Current condition and future directions for. . .	adjacent	Lean construction	Empirical study
9	Digital Transformation of Forest Services in. . .	peripheral	Agile software tools	Case study
10	Direct and configurational paths of open. . .	capability	Organisational agility	Survey/SEM
11	Do vulnerability mitigation strategies. . .	peripheral	Supply-chain risk and resilience	Survey
12	EXPLORING THE IMPACT OF AGILE PROJECT. . .	direct	Agile project management practices	Survey/PLS-SEM
13	Helping Deluxe Beds to sleep easy: A case. . .	direct	Scrum project management	Case study
14	Implementation of an Industry 4.0 Strategy. . .	adjacent	Industry 4.0 and lean	Simulation and case study
15	Information System Development by Using. . .	peripheral	Scrum in service-learning	Educational study
16	Information transparency in SME network. . .	capability	Supply-chain agility	Longitudinal case study
17	Innovation and the SME: Identifying optimal. . .	peripheral	Innovation management briefing	Secondary briefing
18	Innovation, agile project management and. . .	direct	Agile project management	Comparative cases and SEM
19	Intellectual agility and innovation in micro. . .	capability	Intellectual agility	Survey
20	Investing in lean to improve basic. . .	adjacent	Lean production	Longitudinal case study
21	Knowledge, values and. . .	peripheral	Internationalisation and agility mention	Editorial introduction
22	Lean and agile in small- and medium-sized. . .	adjacent	Lean management and supply-chain agility	Comparative cases
23	Linking SME international marketing agility. . .	capability	International marketing agility	Survey
24	Managing Network Resource and Organizational. . .	capability	Organisational agility	Empirical study
25	Multi-level corporate entrepreneurship in. . .	peripheral	Corporate entrepreneurship	Survey/QCA
26	Network resource meets organizational. . .	capability	Organisational agility	Empirical study
27	Optimizing firm's supply chain resilience in. . .	peripheral	Supply-chain resilience	Survey
28	Organic structure and organisational. . .	capability	Workforce agility	Survey/SEM
29	Organisational agility, business best. . .	capability	Organisational agility	Survey/SEM
30	Overall Equipment Efficiency Improvement. . .	adjacent	Lean manufacturing	Case study
31	QCDM culture design: A new paradigm in. . .	peripheral	QCDM quality culture	Model proposal
32	SMART-LOGISTICS FOR PEOPLE MANAGEMENT OF. . .	direct	Agile people-management model	Conceptual model
33	SMEs' sustainable innovation performance. . .	capability	Market-capitalising agility	Survey/PLS
34	Strategic agility-driven business model. . .	capability	Strategic agility	Longitudinal case study
35	Strategic Agility, Exaptation, and Business. . .	capability	Strategic agility	Single case study
36	Strategic Partnership: How Important for. . .	capability	Organisational agility	Survey/SEM
37	Supply chain agility as an acclimatisation. . .	capability	Supply-chain agility	Comparative case study

Table A1. *Cont.*

ID	Study (Short Title)	Class	Focal Construct	Design
38	Supply chain integration and resilience in. . .	capability	Supply-chain agility/resilience	Case study
39	Supply chain management antecedents of. . .	capability	Supply-chain agility	Survey
40	Supply Chain Performance Based on the. . .	direct	Lean-agile operations	Cross-sectional survey/SEM
41	Sustaining innovation during downsizing. . .	capability	Market-capitalising agility	Survey/PLS
42	Textile management enabled by lean thinking:. . .	adjacent	Lean thinking	Case study
43	THE CHANGE IN MANAGEMENT STYLE DURING THE. . .	direct	Agile project management	Survey/Fisher test
44	The Effect of Innovation on the Company's. . .	direct	Lean-agile project management office	Survey/SEM
45	The effect of SMEs' dynamic capability on. . .	capability	Organisational agility	Survey
46	The Role of Supply Chain Resilience on SMEs' . . .	capability	Supply-chain resilience/agility	Empirical study
47	The sequential effect of absorptive. . .	capability	Strategic agility	Empirical study
48	The Urgency of Business Agility During. . .	capability	Business agility	Survey/SEM
49	Exploring agile methods in construction. . .	direct	Agile methods in construction	Multiple case study
50	The co-evolution of IT competence,. . .	capability	Organisational agility	Multiple case study
51	Analysing the impact of agile supply chain. . .	capability	Agile supply chain	Survey/SEM
52	An investigation of agile manufacturing. . .	direct	Agile manufacturing practices	Survey/SEM
53	The relationship between work organization. . .	capability	Workforce agility	Questionnaires

The primary organizational area, historical driver/barrier table links, source abstract, coding notes, record-level eligibility flags, and the complete extraction are available in the accompanying CSV. Historical theme links reproduce the original tables and are not independent validation of each semantic assignment.

Table A2. Study context and principal finding in each original record. Findings are condensed from the supplied PDFs and preserve the original study designs.

ID	Study Country	Industry	SME Context	Principal Finding
1	Not stated clearly	Digital gig economy	One SME digital enterprise	The case reports more projects won and higher turnover after agile deployment, alongside improved engagement and knowledge management.
2	Not stated clearly	Mixed SME digital business	Three SMEs operating for at least ten years	The three cases illustrate a digital development model that emphasizes organizational interactions and communication rather than technology alone.
3	Italy	Fashion apparel	Four companies, including two SMEs and two large firms	The SME cases relied more on industry knowledge and employee intuition, whereas larger firms relied more on technology and formal practices.
4	Indonesia	Home-food and cookie production	One home-food SME case	Waste analysis identified excess transportation; the paper proposes a revised layout and operating procedure for cookie production.
5	India	Pump and foundry manufacturing	68 usable responses from 100 contacted SMEs	Respondents reported awareness of agile manufacturing; the authors call for more effective implementation.

Table A2. Cont.

ID	Study Country	Industry	SME Context	Principal Finding
6	Germany	Mixed SME innovation projects	Five SMEs	The cases motivate combining conventional innovation management for exploitation with agile innovation management for exploration.
7	India	Ceramic manufacturing	Indian ceramic SMEs and industrial experts	Expert ranking places worker training, lean-tool selection, and scrap reduction among the leading proposed success factors.
8	United Kingdom	Highways construction	20 manager interviews followed by 110 questionnaire responses across the highways supply chain	Short-term contracting, fragmentation, and weak business-case communication hinder SME involvement in lean construction.
9	Finland	Forest services	Micro, small and medium forest-service providers and stakeholders	Stakeholders developed and assessed software-service concepts, with benefits left for further evaluation.
10	China	Mixed SMEs	Survey of 245 SMEs	SEM and fsQCA associate combinations of open innovation and organizational agility with business-model innovation.
11	Pakistan	Textile manufacturing	Survey of 355 textile SME entrepreneurs	The survey links vulnerability-mitigation strategies with lower supply-chain risk and better reported firm performance.
12	China	Manufacturing	Survey of 326 production workers from SMBs	PLS-SEM reports positive associations of agile project-management practices with supply-chain resilience and sustainable project management.
13	United Kingdom	Beds and mattress manufacturing	One family-run SME with a technically novice workforce	The case describes a partial Scrum implementation for operational improvement and innovation; it does not provide a comparative effect estimate.
14	Canada	Manufacturing	One Quebec SME in action research and simulation	A tailored strategy was implemented and explored through a single-case action-research and simulation design.
15	Chile	Software-engineering education	One university course with an SME community partner	The course study reports improved teamwork and satisfaction among students and community partners.
16	Japan	Hosiery and fashion supply network	Longitudinal case of one Japanese SME and its network	Information transparency and postponement helped the network respond proactively to market demand.
17	Not applicable	General SME management briefing	Secondary executive briefing, no new SME sample	The briefing summarizes other research and does not report an original agile-method study.
18	China	High-tech SMEs	Two comparative SME cases plus quantitative model testing	Five of six hypotheses were supported; innovation capability and project agility were linked with firm performance under environmental conditions.

Table A2. *Cont.*

ID	Study Country	Industry	SME Context	Principal Finding
19	Serbia	Mixed micro and small businesses	Survey of 110 firms	Intellectual agility was associated with innovativeness, with entrepreneurial leadership reported as a mediator.
20	Sweden	Contract manufacturing	One SME supplier followed longitudinally	Process stability was associated with both cost efficiency and flexibility during the observed change process.
21	Not applicable	International entrepreneurship	Editorial introduction to a special issue	The article introduces a special issue and contains no original SME sample or method-adoption finding.
22	South Africa	Manufacturing and supply chain	Two manufacturing SMEs at different lean-implementation stages	The company further along in lean implementation appeared better positioned for speed and flexibility, though both had stockouts and long lead times.
23	Thailand	Exporting SMEs	Survey of 162 SME exporters	Marketing agility was associated with adoption of internet-enabled inventory systems; perceived advantage increased and complexity reduced adoption.
24	Taiwan	Mixed SMEs and interfirm networks	260 usable SME responses from 400 distributed questionnaires	The article proposes and tests a link between external network resources, organizational capabilities and performance.
25	Spain	Mixed SMEs in metropolitan Madrid	Survey of 233 SME managers	Corporate entrepreneurship depended on combinations of organizational and individual conditions; agile-method training was one variable, not the focal intervention.
26	Taiwan	Mixed SMEs	Survey of 260 SMEs	Network resources and sensing/responding capabilities were associated with performance, with agility included as a mediator.
27	Bangladesh	Mixed industries	Survey of 310 SME managers	Supply-chain capabilities were associated with resilience and reported sustainable performance; resilience mediated the modeled relation.
28	Iran	Mixed SMEs	161 usable SME questionnaires from 360 distributed	Learning, decentralized decisions and flatter structures correlated with workforce agility; low formalization was not reported as significant.
29	South Africa	Mixed SMEs	Cross-sectional survey of 564 owner-managers	Technology capability, collaborative innovation, learning and alignment were associated with agility, which was associated with performance.
30	Egypt	Candle manufacturing	One small candle manufacturer	The case reports increased daily production and reduced cycle time and material waste after the lean intervention.

Table A2. Cont.

ID	Study Country	Industry	SME Context	Principal Finding
31	India	Pump manufacturing	Selected Indian SME context; sampling detail limited	The paper proposes a quality-cost-delivery-morale approach for SMEs; the abstract does not document an agile-method effect.
32	Not stated as an SME study site	Innovative SMEs and urban services	Proposed people-management model with an Erasmus project illustration	The paper proposes a smart-logistics people-management sequence and reports an illustrative project test, without a clearly delimited SME sample.
33	China	Manufacturing	306 top and middle management survey responses	Knowledge coupling was associated with innovation performance, partly through ambidexterity and market-capitalizing agility.
34	Spain	Temporary-work services	One SME temporary-work agency followed longitudinally	Leadership unity and resource fluidity fit the SME case; strategic sensitivity was less natural and resourcefulness emerged as important.
35	Turkey	Insurance services	One Istanbul insurance-agency SME	Strategic sensitivity and resource fluidity together enabled exaptation in the observed business-model innovation process.
36	Not stated clearly	Mixed SMEs	Survey of 135 SME managers; sector and country details limited	Professional knowledge related to agility, and agility related to reputation; partnership did not directly relate to agility.
37	United Kingdom	Mixed SME supply chains	Comparative cases of three British SMEs	Attitudes, resources, power relations and access to information shaped how the SMEs developed supply-chain agility under uncertainty.
38	China	Pig farming and food supply chain	Four pig-sector supply-chain cases involving small and medium farms	Supply-chain integration was associated with resilience through agility and robustness across the four cases.
39	South Africa	Mixed SMEs	407 owners, managers and professional employees in Gauteng	Four SCM strategies predicted agility; technology adoption was the strongest predictor, and agility predicted supply-chain performance.
40	Indonesia	Garment manufacturing	Owners of 160 garment SMEs in Central Java	Agile supply-chain and partnership constructs related to supply-chain performance, whereas leanness alone did not show the same relation.
41	China	Manufacturing	306 valid responses from SME managers, 157 downsizing and 149 not downsizing	Knowledge coupling related to agility and innovation; agility mediation through digitization appeared in the downsizing subgroup.
42	Sweden	Textile manufacturing and management	Two medium-sized textile firms with roughly 150–200 employees	Lean changes improved coordination and communication in two cases; work stress rose in one and fell in the other.
43	Slovakia and Czechia	Mixed SME projects	119 project-manager survey responses, many from SMEs	Most respondents judged a mid-project change feasible in suitable project areas; no real transition was observed.

Table A2. Cont.

ID	Study Country	Industry	SME Context	Principal Finding
44	Iran	High-tech SMEs	108 firms sampled from a stated population of 150 companies	The modeled lean-agile PMO mediated relations between innovation constructs and reported performance.
45	South Korea	Manufacturing, distribution and other sectors	288 firm representatives in Daegu and Gyeongbuk	Marketing capability related to agility, and agility related to innovation capability; uncertainty moderated one technical-capability path.
46	Saudi Arabia	Mixed SMEs	255 SME managers	Supply-chain agility and flexibility were associated with production and market performance; robustness related only to production performance.
47	Turkey	Family-business SMEs	Survey of 421 family-business SMEs	Absorptive capacity related to strategic agility, which related to competitive advantage and reported sustainable performance.
48	Indonesia	Product and service distribution	202 SME owner or manager responses	Business intelligence and IT capability related to business agility, with organizational learning mediating an IT path.
49	Not stated clearly	Construction	Twelve construction SME cases	Interviewed firms showed some agile-like practices, but adoption of named methods was assessed prospectively rather than observed across all cases.
50	China	E-commerce retail	Ten entrepreneurial e-tailers on a Chinese platform	The cases describe co-evolution of IT competence, organizational agility and entrepreneurial action.
51	Bangladesh	Mixed SME supply chains	250 usable SME supply-chain manager responses from 300 distributed surveys	Alertness, flexibility, decisiveness, swiftness and accessibility related positively to sales performance; technological integration moderated the model.
52	India	Automotive component manufacturing	Auto-ancillary SMEs in Jharkhand and eastern India	SEM supports modeled relations among manufacturing responsiveness, strategy, workforce, technology and management agility.
53	United States	Small manufacturing	Six firms with 50-200 employees; 175 participating managers and workers	Autonomy and cooperative relationships predicted workforce agility; demands and uncertainty also affected it.

The companion CSV also distinguishes observed implementation, survey association, model proposal, and peripheral publication; records enablers and barriers only when identifiable in the study text; and gives a provisional reporting appraisal for each record. Codes Y, P, N, and NA mean clearly reported, partly reported, not reported, and not applicable. The five checks cover SME context, sample or case description, data collection, analysis, and limitations. They are not a validated risk-of-bias score.

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